Project Synopsis

Cipher —

AI-Powered University Student Matching Platform

"Find your crew."

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1. Introduction

Cipher is an AI-powered university student matching platform that connects students based on who they truly are — their goals, skills, hobbies, and intensity. Unlike conventional study-finder tools that match students by subject alone, Cipher decodes identity. Every student on campus has a unique set of ambitions and passions, yet most go through university without ever finding the right people to grow with.

The core problems Cipher solves:

- Students struggle to find others with the same academic goals or project ambitions
- Existing platforms like WhatsApp groups or Facebook are too generic and unstructured
- There is no system that understands student identity beyond their course or batch
- Meaningful collaboration rarely happens because the right people never find each other

Cipher addresses these gaps through an intelligent onboarding experience, semantic vector-based matching, and a structured collaboration layer — all in one platform.

2. Objective

The main objective of Cipher is to build an AI-native platform that enables meaningful student connections at the university level. The specific objectives are:

- Smart Onboarding: Allow students to describe themselves naturally through a conversational AI interface — the Cipher Oracle — which automatically extracts their goals, skills, and interests
- Semantic Matching: Use vector embeddings and pgvector to calculate a compatibility score between students based on intent and personality, not just keywords
- Meaningful Connection: Enable students to send connection requests, message matched peers, and build a personal network on campus
- Crew Collaboration: Allow matched students to form crews, work toward shared goals, and track collective progress through tasks, events, and milestones
- Goal-First Discovery: Prioritize matching by shared goals before secondary factors like hobbies or faculty, ensuring the most relevant connections surface first

3. Scope of the Project

Users of the System:

- University students (primary users)
- Lecturers and staff (secondary — can create profiles and be matched for mentorship)

Functional Coverage:

- AI-powered onboarding that converts natural language into structured profile data
- Semantic profile matching with percentage-based compatibility scores
- Connection and direct messaging between matched users
- Crew creation with collaborative task tracking, event scheduling, and milestone management
- Discovery feed to browse public profiles filtered by goal, skill, or interest
- Real-time notifications for matches, messages, and crew activities

Out of Scope (Phase 1):

- Video or voice calling
- Mobile application (web-only for now)
- Payment or premium features

4. System Modules

1) AI Onboarding — The Cipher Oracle

- New students describe themselves through a natural chat conversation
- Gemini API (Cipher Oracle persona) extracts goals, skills, interests, and intensity level
- Extracted data is saved to the database and a vector embedding is generated from the user summary
- Onboarding completes when the Oracle has enough data — no dropdowns, no checkboxes

2) Matching Engine

- Each user profile is converted into a 768-dimension vector embedding using pgvector
- Matching priority: shared goals (highest) → skills → hobbies → faculty/course
- Compatibility score displayed as a percentage on each match card
- AI-generated shared context explains why two users match (e.g., 'Both want to build a startup using Python')

3) Profile System

- Name, faculty, course, year, profile photo
- Goals, skills, and interests (auto-extracted by Oracle during onboarding)
- Intensity level: Casual / Dedicated / Vanguard
- Visibility toggle: Public / Match-Only / Private

4) Connection & Messaging

- Send and receive connection requests
- Accept, decline, or remove connections
- Direct messaging available only between connected users
- Read receipts and new message notifications

5) Crew & Progress Tracking

- Create a crew with a name, purpose, and description
- Invite matched connections to join as members or admins
- Create and assign tasks with status tracking (To Do / In Progress / Done)
- Schedule crew events with location and date
- Set milestones to track collective progress toward a shared goal

6) Discovery & Notifications

- Browse public student profiles with filters for goal, skill, faculty, and interest
- Search by name or keyword
- Notifications for new matches, connection requests, messages, and crew updates

5. Workflow of the System

Step-by-Step User Journey:

- Student registers using their university email and sets a password
- Cipher Oracle initiates a natural conversation to decode the student's identity
- Oracle extracts goals, skills, and interests — profile is auto-populated and embedding is generated
- Student enters the Discover page and views their top matches ranked by compatibility score
- Student sends a connection request to a matched peer
- Once accepted, both can message each other directly
- Students can form or join a crew, assign tasks, schedule events, and track milestones together
- Notifications keep users updated on all activity across their network

6. Database Design

Core tables and key fields:

Table	Key Fields

users	id (UUID), email, name, faculty, raw_bio, embedding (vector), intensity_level, is_onboarded
user_goals	id, user_id, label_id, visibility_level
user_skills	id, user_id, label_id, visibility_level
user_interests	id, user_id, label_id, visibility_level
matches	id, user_id, matched_user_id, semantic_score, shared_context (JSONB)
connections	id, requester_id, receiver_id, status
messages	id, sender_id, receiver_id, content, is_read
crews	id, name, purpose, created_by
crew_members	id, crew_id, user_id, role
tasks	id, crew_id, title, assigned_to, status, due_date
events	id, crew_id, title, location, event_date
milestones	id, crew_id, title, is_achieved
notifications	id, user_id, type, content, is_read

7. Technologies Used

Layer	Technology	Purpose
Frontend	React.js	User interface and client-side logic
Backend	Node.js + Express.js	REST API and server-side logic
Database	PostgreSQL + pgvector	Data storage and vector-based semantic matching
AI — Onboarding	Gemini API	Natural language extraction during onboarding
AI — Matching	pgvector (vector embeddings)	Semantic similarity scoring between profiles
Authentication	JWT (JSON Web Tokens)	Secure user authentication and session management
DevOps	Docker + GitHub Actions	Containerization and CI/CD pipeline
Deployment	Cloud / VM Server	Hosting and production environment

8. Advantages

- AI-native from day one — no rule-based matching, no keyword filters
- Understands student intent and personality, not just subject or course
- Onboarding feels like a conversation, not a form
- Semantic matching finds students who are a 90% match even if they use different words
- Crew progress tracking turns connections into actual collaboration
- Scalable architecture — new modules like room booking or events can be added later
- Unique in the university space — no existing platform combines AI matching with crew management

9. Conclusion

Cipher is more than a study-partner finder — it is a platform that decodes who a student truly is and connects them with the right people to grow alongside. By combining AI-powered onboarding, semantic vector matching, and structured crew collaboration, Cipher solves a real problem that every university student faces: finding their crew. Built on a modern full-stack architecture with React, Node.js, PostgreSQL, and the Gemini API, Cipher demonstrates both technical depth and genuine real-world value.